
Practical 06 for “Topological data analysis” (DAT.C103UF)

The paper “Extracting insights from the shape of complex data using topology” contains three applications of mapper, two of which were discussed in the course. The second application, the voting behavior of US House of representatives, was not discussed. Discuss the outcomes of this part and try to reproduce the results of this part.

For the reproduction part, you need to download the congressional voting records and probably process the data to get it into a suitable format for mapper. It is allowed to use LLMs for this particular sub-task.

This project can also be done by a team of two people. In that case, try to do the analysis for several years and try to see whether the results have been significantly changed over time.